

3. Development Roadmap – AI-Based Smart Quiz System

This roadmap outlines the structured development plan for the AI-Based Smart Quiz System from Week 06 to Week 11 of the semester. The roadmap adopts a phased approach, whereby each week targets a distinct layer of the system—progressing from backend architecture and database engineering through frontend implementation, AI integration, feature completion, quality assurance, and, finally, deployment. Responsibilities are distributed among team members in alignment with their designated roles, ensuring parallel workstreams where feasible and clear accountability at each stage. The schedule has been designed to accommodate iterative review and testing throughout the development cycle, thereby minimising integration risk ahead of the final submission.

Phased Development Schedule

Week	Module / Task	Team Member(s) Responsible	Key Deliverables
06	Backend Development & Database Architecture	Sudipto Das, Aditi Roy	Laravel 12 project scaffolding; authentication with role-based access control (Teacher/Student); database schema (users, courses, questions, quiz attempts, leaderboard); Eloquent models and migrations; RESTful API routes for core modules.
07	Frontend Implementation	Chaon Das	Responsive Blade/Vue layouts for Teacher and Student dashboards; course listing and management views; MCQ creation form; quiz-taking interface with per-question

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			countdown timer; result and explanation pages; leaderboard display.
08	AI Service Integration – Gemini API	Rakibur Rahman Mishan, Sudipto Das	Google Gemini API client integration; automatic MCQ generation endpoint for Mobile Computing Lab; AI-generated question review and storage workflow; error handling and rate-limit management for the Gemini service.
09	Feature Completion & Cheating Detection	Sudipto Das, Chaon Das	Per-question timer logic (configurable per course); tab-switch detection via JavaScript visibilitychange event; is_suspicious flag logic (average time per question < 5 seconds); Teacher dashboard alerts for flagged attempts; end-to-end feature integration and smoke testing.
10	Testing, Debugging & Quality Assurance	All Members	Unit and feature tests (PHPUnit); frontend usability testing; AI response validation; cheating-detection edge-case testing; bug triage and resolution; performance

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			review; code refactoring and documentation.
11	Deployment & Final Submission	All Members	Production environment setup; database seeding with sample data; final integration testing on live server; project report finalisation; presentation preparation; submission of source code, documentation, and demo.

Detailed Module-Wise Responsibilities

Module	Scope of Work	Assigned Team Member(s)
Backend Logic	Laravel 12 application architecture; MVC implementation; role-based authentication and middleware; RESTful API endpoints for quiz management, attempt recording, and leaderboard aggregation; cheating-detection service logic; timer configuration per course.	Sudipto Das
Frontend & UI/UX	Blade templating and/or Vue.js component design; responsive layouts for Teacher and Student portals; interactive quiz-taking interface with live per-question countdown; result and explanation views; leaderboard table; accessibility and cross-browser compatibility.	Chaon Das

Module	Scope of Work	Assigned Team Member(s)
Database Engineering	Relational schema design for users, courses, questions, options, quiz attempts, and leaderboard tables; Eloquent migrations and seeders; query optimisation; indexing strategy; data integrity constraints; Entity-Relationship (ER) diagram maintenance.	Aditi Roy
Artificial Intelligence	Google Gemini API integration for automated MCQ generation targeting the Mobile Computing Lab course; prompt engineering for question quality; parsing and storing AI-generated content; fallback and validation logic for API responses.	Rakibur Rahman Mishan
Testing & QA	Unit tests for backend services and controllers (PHPUnit); feature tests for quiz flow and cheating detection; frontend usability and regression testing; AI output validation; bug reporting and resolution tracking; final pre-deployment verification.	All Members

Milestone Summary

Milestone	Tentative Completion	Status
UI/UX Design Finalisation	Week 05	☐ Completed
Backend & Database Setup	Week 06	☐ In Progress
Frontend Implementation	Week 07	☐ Pending
AI Integration (Gemini API)	Week 08	☐ Pending

Milestone	Tentative Completion	Status
Feature Completion & Cheating Detection	Week 09	☐ Pending
Testing & Debugging	Week 10	☐ Pending
Deployment & Presentation	Week 11	☐ Pending

Deliverables Overview

The following summarises the primary deliverable(s) expected at the conclusion of each development week:

- Week 06 – Backend & Database: Fully configured Laravel 12 application with role-based authentication, complete database schema, Eloquent models, migrations, and core API routes.
- Week 07 – Frontend: Responsive Teacher and Student dashboards, quiz-taking interface with per-question countdown timer, results page, and leaderboard view.
- Week 08 – AI Integration: Operational Google Gemini API integration enabling automatic MCQ generation for the Mobile Computing Lab course, with validated question storage.
- Week 09 – Feature Completion: Fully functional cheating-detection system (tab-switch tracking, is_suspicious flag), configurable per-question timer, and integrated end-to-end application flow.
- Week 10 – Testing & QA: Comprehensive test suite (unit and feature), resolved bug backlog, performance review report, and refactored, documented codebase.
- Week 11 – Deployment & Submission: Live deployment on production server, seeded demo data, final project report, presentation slides, and complete submission package (source code, documentation, and demo).